GUGULOTHU LINGANNA

Hyderabad, India guguloth.linganna124@gmail.com +91 7993101615 LinkedIn.in
Gugulothu-Linganna

Education

National Institute of Technology (NIT) Srinagar — **Srinagar**
B.Tech in Computer Science and Engineering — *2021 – 2025*
- CGPA: 6.34/10.0

Sri Balaji Junior College (SBJC) Waddepally — **Warangal**
Class 11, 12 — *2017 – 2019*
- Percentage: 91.2/100

Zilla Praja Parishad High School (ZPPHS) Bibinagar — **Yadadri Bhongir**
Class 10 — *2016 – 2017*
- CGPA: 8.8/10.0

Projects

Automatic Signal Strength Adjustment Simulation
- Innovated a system to adjust signal strength based on user activity, optimizing performance and ensuring 100 percent environmental safety while minimizing resource waste.

Designed Networks
- Engineered network infrastructures using optimal topologies (star, bus, mesh, ring, hybrid) to improve performance, reliability, and scalability, enhancing data flow efficiency.

Vulnerability Assessment
- Conducted vulnerability assessments in 50+ systems, identifying 100+ security flaws.
- Prioritized remediation, improving security posture by 83 percent within 6 months.

Password Encryption and Decryption
- Deployed encryption algorithms to secure passwords, ensuring 100 percent data protection and safe decryption for authorized access.

Volcanic Eruption Simulation using C Grphics
- Created a volcanic eruption simulation using Turbo C++ graphics, animating lava flow, ash clouds, and smoke dispersion, achieving 99 percent accuracy in visualizing volcanic dynamics.

Anti-Radar Detection System
- Developed a Python-based Anti-Radar Detection System to identify radar signals based on frequency and strength, with customizable detection thresholds. Optimized signal filtering and real-time detection, improving accuracy by 30 percent.

Finance Data Project
- Conducted a comprehensive exploratory data analysis (EDA) on stock prices for major banks.
- Analyzed stock data spanning from January 2006 to January 2016 focusing on data manipulation techniques. Explored stock trends and financial metrics to derive insights into market performance and investment patterns.

Advancements in Intrusion Detection and Prevention Systems
- Designed an anomaly-based intrusion detection system (IDS) using AI / ML algorithms, achieving a 95 percent accuracy rate in detecting five major cyberattacks, including DoS and DDoS.
- Integrated IDS/IPS with firewalls and honeypots, improving real-time attack detection and prevention by 30 percent, enhancing overall network security performance.

Experience

Ethical Hacking and Penetration Testing Intern (C-DAC) **India**
Ministry of Electronics and IT, Government of India — Noida *Dec 2024 – Jan 2025*

- Enhanced harassment detection through email communication (spoofed email) and the data exchanged within it using the memory dump.

Cybersecurity Intern (AICTE) **India**
Shripriti Educational and IT Hub — Madhya Pradesh *Jan 2024 – Feb 2024*

- Implemented OWASP Risk Calculator, Steganography, and Scapy-based packet analysis, reducing incident response time by 70 percent. Performing vulnerability scans and reverse engineering keyloggers, improving security and reducing risk exposure by 50 percent.

Skills

Programming Languages: C, C++, Python, Java, SQL.

Developer Tools: Git, GitHub, VS Code, CodeBlocks, Dev C++, PyCharm, Jupyter, Eclipse, Turbo C++.

Operating Systems: Android, Windows, Linux (Kali, Ubuntu, Parrot).

Network Protocols and Models: IPv4, IPv6, DHCP, SMTP, DNS, HTTPS, IPSec, TCP/IP, OSI.

Network Design and Simulation Tools: Cisco Packet Tracer, NetSim (Academic version).

Network Security Tools and Frameworks: Nmap, Zenmap, Wireshark, VPNs, MITRE ATTandCK, NIST.

Databases: MySQL, SQLite.

Cloud Platforms: AWS, Google Cloud, Microsoft Azure.

Course Work: DSA, OOPS, DBMS, Operating Systems, Computer Networks, Network Security(Cryptographics), Cloud Computing, Software Engineering.

Microsoft Office tools: Word, Excel, PowerPoint.

Certifications

- Ethical Hacking and Penetration Testing (C-DAC) - Ministry of Electronics and IT, Govt. of India.
- Certified Cybersecurity Technician (CCT) - EC-Council.
- Certified Ethical Hacker (CEH) - EC-Council.
- Cyber Hygiene Practices - Ministry of Electronics and IT (MeitY), Govt. of India.
- Cisco LABS Crash Course - EC-Council.
- Android Bug Bounty Hunting: Hunt Like a Rat - EC-Council.
- Unleashing Git and GitHub (1-Day Workshop) - IET-NIT Srinagar.
- Cybersecurity - Oxford Home Study Center (OHSC).
- Cyber Security and Ethical Hacking - Coincent, UDGAM IIT GUWAHATI.

Achievements and Interests

- Member of the Microsoft 365 Developer Program.
- Solved 500+ coding problems, improving personal coding efficiency by 50 percent.
- Led initiatives as School Pupil Leader (SPL), positively impacting the school community.
- Awarded Student of the Year by TATA Motors for academic excellence.
- Advanced to the 1st level of the Election Commission of India competition and 2nd level of the National Highway Awareness Program.
- Received a 50 percent scholarship for academic excellence in +1 and +2 .
- Interested in cybersecurity and ethical hacking, real- world problem solving, generating innovatives, listening to music and watching science fiction films.